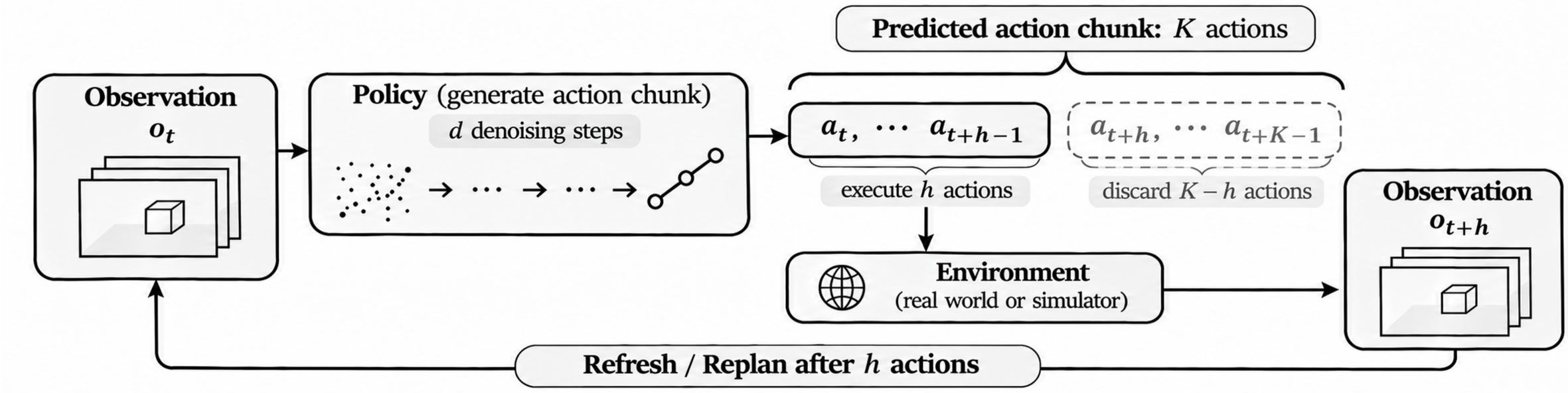
Refinement vs. Refresh

Compute Allocation for Closed-Loop Robot Policies

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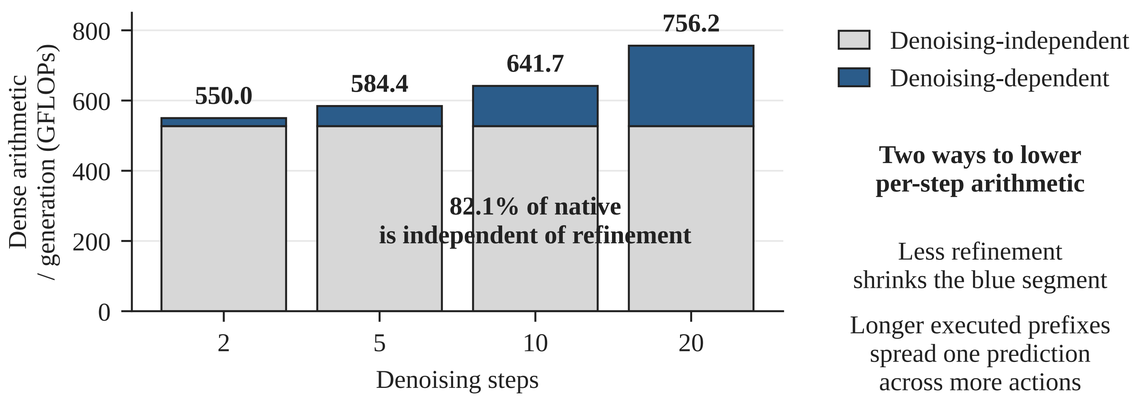
Chunked Closed-Loop Policy



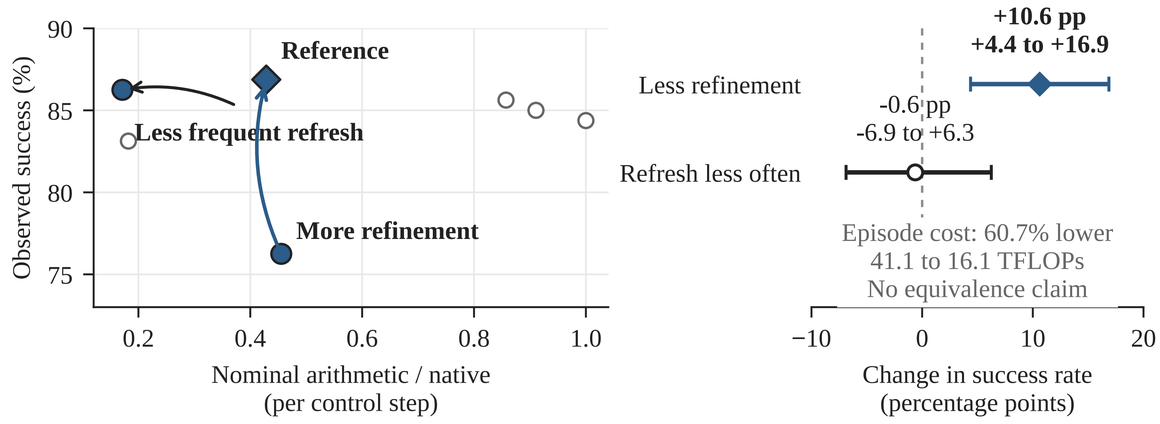
?

Under a fixed inference budget, should we allocate compute to additional denoising steps (**refinement**) or to more frequent observation-conditioned replanning (**refresh**)?

Why the Trade-Off Exists



Held-Out Success and Compute



A Recipe for Closed-Loop Compute Studies

Evaluate allocations within each model

1. Define refinement and refresh controls.
2. Measure the complete cost of a fresh prediction.
3. Freeze primary comparisons before held-out evaluation.
4. Report success and expenditure together.

Interpretation boundary

SmolVLA is the worked example, not a universal allocation rule. Arithmetic is not deployment latency, and similar observed success does not establish equivalent quality.